

Problem

A three-phase generator supplied 10 kVA at a power factor of 0.85 lagging. If 7,500 W are delivered to the load and line losses are 160 W per phase, what are the losses in the generator?

Step-by-step solution

Step 1 of 3

Consider the expression for the power supplied by the generator.

$$P_s = P_g + 3P_l + P_L \dots\dots (1)$$

Here, P_s represents the power supplied by the generator,

P_g represents the power losses in the generator,

$3P_l$ represents the total three line losses,

P_L represents the power delivered to load.

Step 2 of 3

Calculate the average power supplied by the generator, P_s .

$$P_s = S(\text{pf})$$

Substitute 10 kVA for S and 0.85 for pf.

$$\begin{aligned} P_s &= (10 \text{ k})(0.85) \\ &= 8500 \text{ W} \end{aligned}$$

Therefore, the value of the average power supplied by the generator is P_s is, 8500 W.

Step 3 of 3

Recall equation (1).

$$P_s = P_g + 3P_l + P_L$$

Substitute 8500 W for P_s , 7500 W for P_L and 160 W for P_l .

$$8500 = P_g + 3(160) + 7500$$

$$8500 = P_g + 480 + 7500$$

$$P_g = 8500 - 480 - 7500$$

$$P_g = 520 \text{ W}$$

Therefore, the value of the power loss P_g in the generator is 520 W.